

Memory (Group 4)

Author roles

Author roles were classified using the Contributor Role Taxonomy (CRediT; credit.niso.org) as follows:

Martina Iscar: project administration, conceptualization, data curation, formal analysis, methodology, investigation, software, validation and writing review and editing.

Irene Morales: project administration, conceptualization, data curation, software, formal analysis, visualization writing review and editing.

Amalia Varo: project administration, conceptualization, data curation, software, formal analysis, formal analysis, methodology, visualization writing review and editing.

Ferrán Extremera: Project administration, conceptualization, data curation, methodology, formal analysis, software, validation visualization writing review and editing.

Use of Generative AI in This Assignment

This project used generative AI as a collaborative tool. Specifically, we employed ChatGPT and Gemini to develop the R code by giving it specified prompts, clarify statistical interpretations regarding regression outputs, and improve the storytelling flow of our HR recommendations.

Even though these tools provided the foundation for the code, human intervention was present at every stage. We frequently modified and corrected overcomplicated code and addressed false interpretations—particularly regarding the counter-intuitive relationship between job strain and performance. We understood the nuances of the manufacturing and financial databases much more clearly than the AI. Therefore, we were the primary decision-makers, remaining responsible for the content, explanations, and strategic conclusions. AI was a tool for enhancement, not a replacement for our critical thinking.

Critical Reflection:

In this final case we took all the good practices that enabled the AI to give us good code and interpretations. We found out that giving the data in text and not in images makes the algorithm better analyse and interpret the prompts we provide. Also, even though we split the prompts between each team member we all gave similar prompts at the beginning. We give the whole info of the case; with all the prompts

we aim to analyse and the variable key of the data set. Then each made more personalised prompts to better target their deliverables.

So, our steps were the following:

1. Context and variables. Do not ask for code rightaway but to understand the information provided.
2. Provide with the code
3. Personalised prompts for the elaboration of each prompt.

This allowed us to anticipate the results before running the models in R. This helped develop good initial codes.

With our active role in "prompt engineering" we ensured the AI didn't confuse variables.

Finally, we used AI as a translator and tone-checker. Some of our strategic recommendations were drafted in Spanish to ensure the HR logic was sound, then translated into professional English, ensuring the delivery was appropriate for an executive-level report.

Overall, AI did provide powerful insights and helped us deepen our understanding of the case. Nevertheless, it is our responsibility to proofread the answers given and correct where necessary.

Examples of Significant AI Prompt–Response Interactions

Example 1: Understanding the case as a whole story

- **Prompt:** Please give an overview of the story of the prompts and what the objective and what questions each are posing. This will serve to start the case with a full understanding. *(Note: We gave it the Case 5 files)*
- **Response:** It first gave us the big picture of the case. It then told us that there was a clear narrative dividing the prompts into two phases: front end of hiring (prompts 1-3) and back end of hiring quality (prompts 4-5). It then gave us a clearer view of each individual prompt, briefly explaining the technical tools to use and phrasing the prompts into questions so we could understand the direct message we should give in our deliverables. Lastly it gave us a list of the strategic questions behind the whole case to summarise the whole case.
- **Evaluation:** This was very useful in order to have some context before starting the case and so the whole group had some understanding of everything before we divided the prompts to separately work on them. AI showed it is good at evaluating information and explaining in a more low level way.

Example 2: Exploring Diversity in the Recruitment Pipeline (Prompt 1)

- **Prompt:** Analyse the applicant pool and identify where female or UG candidates are being lost across recruitment stages.
- **Response:** The analysis reveals a clear gender imbalance at the application stage, with fewer female candidates, while UG representation is relatively balanced. As candidates move through the process, there is a strong reduction in numbers during shortlisting and before the interview phase. This indicates that the main filtering occurs early in the pipeline, where diversity is most likely being affected.
- **Evaluation:** This interpretation suggests that potential inequalities are more likely linked to early screening decisions rather than later stages. As a result, organizations should focus on improving and standardizing shortlisting criteria to reduce subjectivity and ensure a fairer recruitment process.

Example 3: Analysing Sickness Absence Limitations (Prompt 7)

- **Prompt:** Predict SickDays2014 using attitude variables. If the results are non-significant, explain why this happens and what other data HR should collect to better understand absenteeism.
- **Response:** It showed that attitudes like "Job Satisfaction" had almost zero impact on sick days. It suggested that sickness is often "Stochastic" (random) or driven by physical health data which HR doesn't usually have for privacy reasons.
- **Evaluation:** This taught us that HR shouldn't blame "low morale" for every sick day. We adjusted our recommendation to suggest "Wellness and Preventive Health" programs instead of just "Engagement" surveys to solve absenteeism.

Example 4: Analysing Recruitment Bias (Prompt 2)

- **Prompt:** Test for bias in the recruitment pipeline (Shortlisting and Offer stages) based on Gender and Underrepresented Group (UG) status using cross-tabulation and chi-square tests. Summarize the findings and determine if the hiring decisions are fair and unbiased.
- **Response:** It showed severe, statistically significant bias at the Shortlisting stage, creating a massive "diversity bottleneck" where women and UG candidates were disproportionately rejected. It also revealed that while gender bias favoring men persists into the final Offer stage, UG finalists who actually make it to this phase are evaluated fairly with no systemic bias.
- **Evaluation:** This taught us that the overall recruitment pipeline is not impartial. We adjusted our focus to urgently audit and recalibrate the initial

shortlisting criteria and algorithms, recognizing that we are losing the vast majority of our diverse talent at the very first screening, long before final interviews take place.